

Continuous.

Let  $f: X \rightarrow Y$  be a map where  $X, Y$  are metric spaces, and  $a \in X$ . TFAE:

a)  $f$  is continuous at  $a$

b) If a sequence  $\{x_n\} \in X$  converges to  $a$ , then  $\{f(x_n)\}$  converges to  $f(a)$ .

Proof. Suppose  $f$  is continuous at  $a$ . Let  $\epsilon > 0$  be given. Then, there exists  $\delta > 0$  s.t.

$$x \in B_\delta(a) \Rightarrow f(x) \in B_\epsilon(f(a))$$

by definition. pick  $N \in \mathbb{N}$  s.t.

$$n \geq N \Rightarrow x_n \in B_\delta(a)$$

Now, for  $n \geq N$ ,

$$f(x_n) \in B_\epsilon(f(a))$$

T. show the converse, using contraposition. Suppose that  $f$  is not continuous at  $a$ . That is, there exists  $\epsilon > 0$  st for any  $\delta > 0$ , there is  $x \in B_\delta(a)$  satisfying

$$d(f(x), f(a)) \geq \epsilon.$$

Therefore, for each  $n \in \mathbb{N}$ , we can choose  $x_n \in B_{\frac{1}{n}}(a)$  and  $d(f(x_n), f(a)) \geq \epsilon$ .

By construction,  $\{x_n\}$  converges to  $a$  but  $f(x_n)$  does not converge to  $f(a)$ .

Thomae's function.

define a map

$$T: (0, \infty) \rightarrow \mathbb{R}: x \mapsto \begin{cases} \frac{1}{p} & x = \frac{p}{q}, \text{ irreducible form (or } \gcd(p, q) = 1) \\ 0 & x \notin \mathbb{Q} \end{cases}$$

Then  $T$  is continuous at every irrational number, but not continuous at rationals.

Proof. Let  $c \in (0, \infty)$  be an irrational number. Given  $\epsilon > 0$ , there is  $N \in \mathbb{N}$  s.t.  $\frac{1}{N} < \epsilon$ .

Since there exist only finite number rationals such that the denominator smaller than  $N$ ,

$$\text{more specifically, } S = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \dots, \frac{N-1}{N}\right\} \left(\text{at most } \frac{N(N-1)}{2}\right)$$

we can choose small enough  $\delta > 0$  s.t.  $(c - \delta, c + \delta) \cap S = \emptyset$ . Now,

$$|x - c| < \delta \Rightarrow |f(x) - f(c)| < \frac{1}{n} < \epsilon$$

Therefore,  $f$  is continuous at irrationals. But, let  $x \in (0, \infty)$  be a rational.

Denote  $x = \frac{p}{q}$  where  $p, q \in \mathbb{N}$ ,  $\gcd(p, q) = 1$ . Taking  $\epsilon = \frac{1}{p}$ . Then the density of irrationals gives

$$\forall \delta > 0, \exists \text{ irrational } t \in (x - \delta, x + \delta) \text{ s.t. } |f(t) - f(x)| = \frac{1}{p}$$

Thus, not continuous at rationals.

Exercise.

20.E Define a map

$$f: \mathbb{R} \rightarrow \mathbb{R} : x \mapsto \begin{cases} x & \text{if } x \notin \mathbb{Q} \\ 1-x & \text{if } x \in \mathbb{Q} \end{cases}$$

Then  $f$  is continuous only at  $x = \frac{1}{2}$ .

Sol. Let  $\varepsilon > 0$  be given. Taking  $\delta = \varepsilon$ . Then for any  $x \in (\frac{1}{2} - \delta, \frac{1}{2} + \delta)$ ,

$$|f(x) - \frac{1}{2}| = \begin{cases} |x - \frac{1}{2}| & \text{if } x \notin \mathbb{Q} \\ |1-x - \frac{1}{2}| & \text{if } x \in \mathbb{Q} \end{cases}$$

Both case satisfies  $|f(x) - \frac{1}{2}| < \varepsilon$ . Therefore continuous.

Let  $x \in \mathbb{R}$  with  $x \neq \frac{1}{2}$ . If  $x \in \mathbb{Q}$ , then by density of rationals, we can construct  $\{x_n\}$  in  $\mathbb{R} \setminus \mathbb{Q}$  st  $x_n \rightarrow x$  as  $n \rightarrow \infty$

But  $f(x_n) = x_n \rightarrow x$  as  $n \rightarrow \infty$ , not converges to  $f(x) = 1-x$ .

Similarly, if  $x \notin \mathbb{Q}$ , it can be shown by the same process.

20.F If a Continuous map  $f: \mathbb{R} \rightarrow \mathbb{R}$  satisfies

$$f(x) = 0 \text{ for all } x \in \mathbb{Q}$$

Then  $f$  is zero map.

Proof. Suppose that  $f$  is not zero map. Given  $x \in \mathbb{R}$  satisfying  $f(x) \neq 0$ , take  $\varepsilon = |f(x)|$ . For any  $\delta > 0$ , by the density of rationals,

$$\leadsto \exists t \in (x - \delta, x + \delta) \text{ st } |f(t) - f(x)| = |f(t)| > \frac{1}{2}|f(x)|.$$

It is contradiction.

20.G Let  $f, g: \mathbb{R} \rightarrow \mathbb{R}$  be continuous map. If  $f(x) = g(x) \forall x \in \mathbb{Q}$ , then  $f = g$ .

Proof. Take  $h = f - g$  and apply 20.F.

20.11. Using  $|\sin x| \leq |x|$ , show sine function is continuous at  $x=0$ .  
Moreover, using

$$\sin x - \sin u = 2 \sin \frac{1}{2}(x-u) \cos \frac{1}{2}(x+u)$$

Show sine function continuous on  $\mathbb{R}$ .

proof. Given  $\epsilon > 0$ , take  $\delta = \epsilon$ . Then,

$$|x| < \delta \Rightarrow |\sin x| \leq |x| < \delta = \epsilon$$

Thus, continuous at  $x=0$ . Similarly, taking  $\delta = \epsilon$  then

$$|x-u| < \delta \Rightarrow |\sin x - \sin u| = \left| 2 \sin \frac{1}{2}(x-u) \cos \frac{1}{2}(x+u) \right| \leq |x-u| < \epsilon.$$

20.12. Define a map

$$g: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \begin{cases} x \cdot \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Then  $g$  is continuous on  $\mathbb{R}$ .

proof. Since  $x$ ,  $\sin x$ ,  $\frac{1}{x}$  are continuous if  $x \neq 0$ , so is  $x \sin \frac{1}{x}$ .

Given  $\epsilon > 0$ , pick  $\delta = \epsilon$ . Then

$$0 < |x| < \delta \Rightarrow \left| x \sin\left(\frac{1}{x}\right) \right| \leq |x| < \epsilon$$

Therefore,  $g$  is continuous everywhere.

20.13. Define a map

$$F: \mathbb{R}^2 \rightarrow \mathbb{R}; (x,y) \mapsto \begin{cases} x^2 + y^2 & \text{if } (x,y) \in \mathbb{Q} \times \mathbb{Q} \\ 0 & \text{otherwise} \end{cases}$$

Determine the points where  $F$  is continuous.

Sol) Consider  $(0,0)$ . Given  $\epsilon > 0$ , taking  $\delta = \sqrt{\epsilon}$

$$(x,y) \in B_\delta(0,0) \Rightarrow |f(x,y)| = x^2 + y^2 < \delta^2 = \epsilon.$$

Thus conti. at zero. But, if  $(x,y) \neq 0$ , regardless of whether  $(x,y) \in \mathbb{Q} \times \mathbb{Q}$  or not,

we can construct a sequence  $(x_n, y_n) \rightarrow (x,y)$  but does not  $f(x_n, y_n) \rightarrow f(x,y)$ .

20. L. A map  $f: \mathbb{R} \rightarrow \mathbb{R}$  is called additive if:

$$\forall x, y \in \mathbb{R}, \quad f(x+y) = f(x) + f(y).$$

$$f(2) = f(1+1) = 2f(1) \quad f\left(\frac{1}{n}\right) = \frac{1}{n}f(1)$$

If additive map is continuous at  $x=0$ , then it is continuous at every point.

Moreover, monotone additive map is continuous everywhere.

Sol. Given  $\varepsilon > 0$ , there is  $\delta > 0$  s.t.

$$|t| < \delta \Rightarrow |f(t)| < \varepsilon \quad (f(0) = 0, \text{ because } f(0) = 2f(0))$$

Since  $f$  is continuous at  $x=0$ . Now, given  $x \in \mathbb{R}$ ,

$$|x-t| < \delta \Rightarrow |f(x) - f(t)| = |f(x-t)| < \varepsilon \quad (\text{since } f(x-t) + f(t) = f(x))$$

by additivity.

Meanwhile, suppose that  $f$  increases monotonically, and additive.

Given  $\varepsilon > 0$ , take  $C \neq 0$  s.t.  $f(C) \neq 0$  (If no such  $C$  exists, then  $f=0$ , there is nothing to prove.)

Observe that  $f(t) = f(\frac{1}{2}t) + f(\frac{1}{2}t) = 2f(\frac{1}{2}t)$  for any  $t \in \mathbb{R}$ . Inductively, for all  $n \in \mathbb{N}$ ,

$$\frac{1}{2^n} f(t) = f\left(\frac{1}{2^n} t\right)$$

Take  $N \in \mathbb{N}$  satisfying  $\frac{1}{2^N} f(|C|) < \varepsilon$ . That is,  $\frac{1}{2^N} f(|C|) = f\left(\frac{1}{2^N} |C|\right) < \varepsilon$

$$\text{Now, } |x| < \frac{1}{2^N} |C| \Rightarrow |f(x)| = f(|x|) < f\left(\frac{1}{2^N} |C|\right) < \varepsilon$$

$\uparrow$   $\uparrow$   
 $0 = f(x-x) = f(x) + f(-x)$   $f$  increases

Therefore  $f$  is continuous at  $x=0$ , so is continuous everywhere.

21. M. Additive continuous map  $f: \mathbb{R} \rightarrow \mathbb{R}$  must be  $x \mapsto f(1)x$ .

Define a map  $\Gamma: \mathbb{R} \rightarrow \mathbb{Z}; x \mapsto \min\{n \in \mathbb{Z} \mid n \geq x\}$  (It is well-defined by well-ordering principle)

And observe that for any  $x \in \mathbb{R}$ ,

$$\lim_{n \rightarrow \infty} \frac{\lceil 2^n x \rceil}{2^n} = x \quad \text{because} \quad 2^n x \leq \lceil 2^n x \rceil < 2^n x + 1.$$

Given  $x \in \mathbb{R}$ ,  $f\left(\frac{\lceil 2^n x \rceil}{2^n}\right) = \frac{\lceil 2^n x \rceil}{2^n} \cdot f(1)$  by additivity. Now,

$$cx = f(1) \cdot x = f(1) \cdot \lim_{n \rightarrow \infty} \frac{\lceil 2^n x \rceil}{2^n} = \lim_{n \rightarrow \infty} f\left(\frac{\lceil 2^n x \rceil}{2^n}\right) = f\left(\lim_{n \rightarrow \infty} \frac{\lceil 2^n x \rceil}{2^n}\right) = f(x)$$

$\uparrow$

I have dyadic rational numbers.  $f$  conti.

20.11 If  $g: \mathbb{R} \rightarrow \mathbb{R}$  satisfies that

$$g(x+y) = g(x) \cdot g(y) \text{ for all } x, y \in \mathbb{R}$$

then  $g$  has the following properties:

1). If  $g$  is continuous at  $x=0$ , then it is continuous everywhere

2). If  $g(x)=0$  for some  $x \in \mathbb{R}$ , then  $g=0$ .

Proof If  $g(x)=0$  for some  $x \in \mathbb{R}$ , then for any  $y \in \mathbb{R}$ ,

$$g(y) = g(y-x+x) = g(y-x)g(x) = 0.$$

The zero map is clearly continuous, suppose that  $g \neq 0 \forall x \in \mathbb{R}$ .

Let  $a \in \mathbb{R}$  be given. Given  $\epsilon > 0$ , there exists  $\delta > 0$  s.t.

$$|t| < \delta \Rightarrow |g(t) - g(0)| = |g(t) - 1| < \frac{\epsilon}{|g(a)|} \quad (g(0+a) = g(0)^2 \text{ implies } g(0) = 1)$$

$$\text{Now, } |x-a| < \delta \Rightarrow |g(x) - g(a)| = |g(a)| \cdot |g(x-a) - 1| < \epsilon.$$

$$\uparrow$$

$$g(x) = g(x-a+a) = g(x-a) \cdot g(a)$$

Doodle:

Observe that  $g(0+a) = g(a) \cdot g(0)$  implies  $g(0) = 0$  or  $1$ .

If  $g(0)=0$ , then for any  $x \in \mathbb{R}$ ,  $g(x) = g(x+0) = g(x)g(0) = 0$ .

Clearly zero map is continuous, suppose that  $g(0)=1$ .

Given  $\epsilon > 0$ , there exists  $\delta > 0$  s.t.

$$|t| < \delta \Rightarrow |g(t) - g(0)| = |g(t) - 1| < \epsilon$$

Since the assumption of 1).

Given  $\epsilon > 0$ , there is  $\delta > 0$  s.t.

$$|t| < \delta \Rightarrow |g(t) - g(0)| < \epsilon.$$

Now, let  $a \in \mathbb{R}$  be given. Then  $g(x) = g(x-a+a) = g(x-a)g(a)$

$$|x-a| < \delta \Rightarrow |g(x) - g(a)| = |g(a)| \cdot |g(x-a) - 1| \leq |g(x-a) - g(0)| < \epsilon$$

$$\uparrow$$

$$g(0+a) = g(a) \Rightarrow g(0) = 0 \text{ or } 1$$

if  $g(0)=0$ , then  $g(x) = g(x+0) = 0$

if  $g(0)=1$ ,

20.0 False. Ask a toddler on the street.

But the converse is true, by the triangle inequality.

20.P. Let  $f, g: \mathbb{R}^p \rightarrow \mathbb{R}$  be continuous at  $a \in \mathbb{R}^p$ , and define

$$h: \mathbb{R}^p \rightarrow \mathbb{R}: x \mapsto \sup\{f(x), g(x)\} \quad \text{and} \quad k: \mathbb{R}^p \rightarrow \mathbb{R}: x \mapsto \inf\{f(x), g(x)\}$$

Prove  $h, k$  are continuous at  $x=a$ .

Proof. Since  $\sup\{f(x), g(x)\} = \frac{1}{2} \cdot (f(x) + g(x)) + |f(x) - g(x)|$

directly  $h$  is continuous at  $a$ , being compositions of continuous.

20.Q Ask...

monotone

20.R. If  $f: I \rightarrow \mathbb{R}$  is <sup>monotone</sup> increasing, then TFAE:

a).  $f$  is continuous at  $c \in I^\circ$

b)  $\forall \varepsilon > 0, \exists x_1, x_2 \in I$  with  $x_1 < c < x_2$  s.t.  $f(x_2) - f(x_1) < \varepsilon$ .

c)  $\sup_{x < c} f(x) = f(c) = \inf_{x > c} f(x)$

Proof. Suppose  $f$  is continuous at  $c \in I^\circ$ . Given  $\varepsilon > 0$ , there is  $\delta > 0$  s.t.

$$|x - c| < \delta \Rightarrow |f(x) - f(c)| < \frac{\varepsilon}{2}$$

Take  $x_1 = c - \frac{\delta}{2}$ ,  $x_2 = c + \frac{\delta}{2}$ . Then clearly  $|x_i - c| = \frac{\delta}{2} < \delta$  ( $i=1,2$ ), and

$$f(x_2) - f(x_1) = |f(x_2) - f(x_1)| \leq |f(x_2) - f(c)| + |f(c) - f(x_1)| < \varepsilon.$$

Conversely, take  $\delta = \min\{|x_2 - c|, |c - x_1|\}$ , then

$$|x - c| < \delta \Rightarrow |f(x) - f(c)| \leq f(x_2) - f(x_1) < \varepsilon.$$

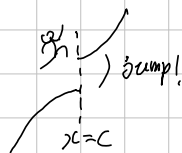
Thus a)  $\Leftrightarrow$  b). And, b)  $\Leftrightarrow$  c) can be derived by directly:

$$\sup_{x < c} f(x) = f(c) \text{ iff } \forall \varepsilon > 0, \exists x < c \text{ s.t. } f(c) - \varepsilon < f(x) \leq f(c)$$

20.5. Let  $f: [a, b] \rightarrow \mathbb{R}$  be monotonically increasing.

Define a map

$$j_c: (a, b) \rightarrow \mathbb{R}; c \mapsto \inf_{x > c} f(x) - \sup_{x < c} f(x)$$



If  $j_c > 0$ , we say that  $f$  has a jump at the point  $c$ .

a) For any  $n \in \mathbb{N}$ , there exist at most finitely many points at which  $f$  has a jump exceeding  $\frac{1}{n}$ .

More set-theoretically, the following set is finite for all  $n \in \mathbb{N}$ .

$$D_n = \left\{ c \in (a, b) \mid j_c > \frac{1}{n} \right\}$$

b). Deduce that monotone function has at most countably many discontinuous points.

Proof. Let  $n \in \mathbb{N}$  be given. Suppose that  $D_n$  has infinitely many elements.

Take  $m \in \mathbb{N}$  satisfying  $n \cdot (f(b) - f(a)) < m$ .

Choose  $c_1, c_2, \dots, c_m \in D_n$  satisfying  $a = c_0 < c_1 < c_2 < \dots < c_m < c_{m+1} = b$ . Then

$$\sum_{i=1}^m j_{c_i} > \frac{m}{n} > f(b) - f(a)$$

Take  $\varepsilon > 0$  s.t.  $\frac{m}{n} - \frac{m}{n} > \varepsilon$ , and put  $r = \frac{1}{2} \cdot \min \{ |c_i - c_{i-1}| \mid 0 \leq i \leq m+1, i \neq 0 \}$ .

For each  $1 \leq i \leq m$ , there exists  $\delta_i > 0$  s.t.

$$\sup_{x < c_i} f(x) - \frac{\varepsilon}{2m} < f(c_i - \delta_i) \leq \sup_{x < c_i} f(x) \quad \text{and} \quad \inf_{x > c_i} f(x) \leq f(c_i + \delta_i) < \inf_{x > c_i} f(x) + \frac{\varepsilon}{2m}$$

That is,

$$j_{c_i} = \inf_{x > c_i} f(x) - \sup_{x < c_i} f(x) \leq f(c_i + \delta_i) - f(c_i - \delta_i) < \inf_{x > c_i} f(x) - \sup_{x < c_i} f(x) + \frac{\varepsilon}{m} = j_{c_i} + \frac{\varepsilon}{m}$$

Finally,

$$\sum_{i=1}^m (f(c_i + \delta_i) - f(c_i - \delta_i)) = \sum_{i=1}^m j_{c_i} + \varepsilon > \frac{m}{n} > f(b) - f(a)$$

But, it is contradiction, because

$$\begin{aligned} f(b) - f(a) &= f(b) - f(c_m + \delta_m) + \sum_{i=1}^m [f(c_i + \delta_i) - f(c_{i-1} + \delta_{i-1})] + f(c_0 + \delta_0) - f(a) \\ &\geq \sum_{i=1}^m (f(c_i + \delta_i) - f(c_i - \delta_i)) \quad (c_i - c_{i-1} \geq 2r > \delta_i + \delta_{i-1}) \end{aligned}$$

Now, we can prove that b). Since  $f$  is continuous at  $c \in (a, b)$  iff  $j_c = 0$ ,

$$\left\{ c \in (a, b) \mid j_c \neq 0 \right\} = \bigcup_{n \in \mathbb{N}} D_n \quad \text{is at most countable.}$$

Note: Since  $\mathbb{R} = \bigcup_{n \in \mathbb{Z}} [n, n+1]$ , we can extend this statement with respect to  $\mathbb{R}$ .

## Section 22.

Topologically, a Continuous map preserves Connectedness and Compactness.

This follows several properties of real-valued function  $f: X \rightarrow \mathbb{R}$

1) (IVT) If  $f: X \rightarrow \mathbb{R}$  is continuous and  $K \in \mathbb{R}$  satisfies  $\inf_{x \in X} f(x) < K < \sup_{x \in X} f(x)$ , then there exists a  $t \in I$  s.t.  $K = f(t)$ .

2) (Max-min Thm) If  $X \subset \mathbb{R}^n$  is compact, then there exists  $x^*, x_* \in X$  such that

$$f(x^*) = \sup_{x \in X} f(x) \quad \text{and} \quad f(x_*) = \inf_{x \in X} f(x)$$